



THE UNIVERSITY OF
NOTRE DAME
A U S T R A L I A

MED 4000
WEEK 4.06

TUTOR GUIDE

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RELEVANT PRIOR LEARNING

MED 1000

1.01 Cardiovascular physiology [L & P]

1.02 Circulatory shock [L & P]

1.22 Renal physiology [L & P]

1.22 The critically ill patient [L]

MED 2000

2.08 Pathophysiology of cardiac failure [L]

2.27 Renal physiology & renal failure [L]

MED 3000

3.01 Cardiac failure [Short case tutorial & L]

3.14 Acute renal failure [short case tutorial]

3.23 Acute liver failure [short case tutorial]

WEEK 06_INTENSIVE CARE MEDICINE		
Yr	Wk	LEARNING OBJECTIVES
4	6	- Outline the key clinical features of acute heart failure and be able to classify the causes. - Outline the principles of management of acute heart failure. - Review cardiovascular physiology in particular the effects of pre-load, after-load, rate & rhythm and contractility on cardiac output. - Explain the role of thrombolysis, stenting & CABG in the management of acute heart failure. - Explain the indications and contraindications of Beta-blockers, ACE inhibitors, diuretics and glycerol trinitrate in the management of acute heart failure. - Outline the mechanism of action, indications and contraindications of the following inotrope classes in the management of acute heart failure; catecholamines [eg noradrenalin], phosphodiesterase inhibitors [eg milrinone], cardiac glycosides [eg digoxin], calcium and calcium sensitisers [eg levosimendan]. - Outline the techniques used in mechanical support of the circulation.
4	6	- Define septic shock. - Outline the clinical features of septic shock and explain the underlying pathophysiology. - Outline the principles of assessment and management of septic shock under the following headings; initial resuscitation, diagnosis, supportive treatment and specific treatment. - In a table overview the following catecholamines – Adrenalin, Noradrenalin, Dobutamine and Dopamine under the headings effect on Alpha1, Beta1, Beta2 and dopamine receptors, effect on cardiac output and systemic vascular resistance.
4	6	- Define acute renal failure - Outline the key causes of acute renal failure and be able to explain the underlying pathophysiology - Outline the key clinical findings in acute renal failure and explain the underlying pathophysiology. - Outline a systematic approach to the assessment of the oliguric patient. - Know the investigations undertaken in the assessment of the patient with possible acute renal failure and describe the indicative results. - Outline the principles of management of the patient with acute renal failure including the indications for dialysis. - Explain the principles of the renal dialysis and outline the main modes of dialysis.
4	6	- Know the key components of the assessment of a patient following a suspected overdose or poisoning and be able to implement these in an emergency situation. - Outline the principles of management of a patient following a suspected overdose. - Outline the assessment and management of patients presenting with paracetamol overdose.

WEEK 06_INTENSIVE CARE MEDICINE		
Yr	Wk	LEARNING OBJECTIVES
4	6	• Define acute hepatic failure. [SDL] • Know the causes of acute liver failure and explain the underlying pathophysiology of the key features of the clinical presentation. [SDL] • Review the physiology of the liver and be able to explain the clinical consequences of derangements in these functions as a result of acute hepatic failure. [SDL] • Outline the principles of management of acute hepatic failure under the headings, initial resuscitation, diagnosis, supportive treatment, specific treatment. [SDL] • Review and summarise the indications for liver transplantation. [SDL]

CASE ONE

Short case number: 4_06_01

Category: Cardiovascular System

Discipline: Intensive Care

Setting: Intensive Care Unit

Topic: Acute heart failure_Cardiogenic shock

CASE



A previously well, 30 yo man presents through ED with extreme fatigue. Several weeks ago he had “the flu”.

He is on no medications and is a non-smoker.

On examination he has a BP of 80/50; pulse of 120/min. He is cold peripherally. JVP to angle of jaw. Fine crackles throughout both lung fields. S3 gallop.

He prefers to sit rather than lie down because he finds it easier to breathe.

Echocardiogram shows severe global impairment of ventricular function. Estimated EF 15%.

QUESTIONS

1. In assessing this patient what are causes of acute heart failure that you consider and what are the typical clinical features seen?
2. Outline the pharmacological management of severe acute heart failure and the role of each medication.
3. Outline the effects of the key classes of medication used in the management of acute heart failure on cardiovascular physiology. In particular consider effects on pre-load, after-load, heart rate, heart rhythm and contractility.
4. As the patient deteriorates mechanical support of his circulation is considered. Describe techniques used for mechanical support of the circulation.

Resources:

- Berston, AD, & Soni, N (2009). Oh's intensive care manual (6th ed.). Philadelphia: Elsevier.
- Cameron et al (2009). Textbook of adult emergency medicine (3rd ed.). Sydney: Churchill Livingstone.

ANSWERS

What are causes of acute heart failure that you consider and what are the typical clinical features seen?

Acute heart failure is not a pathological diagnosis but a collective term for all kinds of myocardial failure. Commonest cause is **coronary artery disease leading to myocardial infarction** especially LAD stenosis, but important to consider other causes. Remember that the normal cardiac function is dependent on preload, afterload, contractility, heart rate and rhythm, and an unobstructed circulation. Preload is impaired with tamponade and mitral stenosis (eg acutely from a thrombosed prosthesis) as well as tension pneumothorax (impaired R sided venous return, as well as afterload increase) (as well as hypovolaemia of course); afterload from severe hypertension eg cocaine overdose, aortic stenosis.

Contractility from ischaemia but also myocarditis and severe cardiomyopathy, also post-CPB in a subgroup of patients; myocardial contusion after blunt trauma. Severe sepsis also impairs myocardial contractility. Severe bradycardia eg CHB reduces cardiac output severely; extreme tachycardia limits filling time and if there is any impediment to ejection eg aortic stenosis this effect is magnified. Major PE both causes acute RV failure and reduces LV filling; Acute MR from papillary muscle rupture, or VSD from septal infarct, or severe acute AR from dissection or endocarditis. Consider overdose of Ca-channel blockers or beta-blockers if appropriate.

Diagnosis depends on history, exam, ECG and particularly echo which will both give a diagnosis in most cases as well as show response to treatment. Other tests include Troponin (rises within 4/24; peak 12-24 hours; detectable for 7-10 days). BNP role uncertain but sensitive and specific as a test for heart failure as a cause of dyspnoea.

Clinical presentation is classically with hypotension, poor peripheral perfusion, tachycardia, raised JVP, decreased mentation, oliguria, orthopnoea, s3 gallop, murmur if a valvular abnormality causing or contributing, tracheal shift with tension PTX, muffled heart sounds with tamponade.

Outline the pharmacological management of severe acute heart failure and the role of each medication.

Drug management involves optimisation of preload and afterload of both LV and RV, contractility, rate and rhythm. In acute Myocardial infarction, therapies to improve myocardial salvage are vital. These include thrombolysis, stenting, CABG, and Beta-blockers if not contra-indicated (hypotension, asthma, heart block); ACE inhibitors (not with hypotension or renal impairment). Used in conjunction with invasive haemodynamic monitoring of BP, CVP, Cardiac output, PAOP. No RCT has proven benefit of PA catheter but widely used and intuitively make sense to use.

Outline the effects of the key classes of medication used in the management of acute heart failure on cardiovascular physiology. In particular consider effects on pre-load, after-load, heart rate, heart rhythm and contractility.

- i. Preload. Decrease with GTN; diuretics; dialysis. May respond to cautious fluid loading. Depends on position on “Starling curve”. If tamponade present, no amount of manipulation will fix the problem without drainage.
- ii. Afterload reduction will allow greater CO. Failing heart very afterload-sensitive. Care not to reduce BP to a level too low for diastolic coronary perfusion. Can use a vasodilator eg SNP but more commonly an “ino-dilator” eg dobutamine or milrinone.
- iii. Rate and rhythm. HR 90/min optimal. Sinus rhythm – treat AFib with rate control (Digoxin) or reversion (electrical or amiodarone, but beware of excessive hypotension). Sinus tachy – ensure analgesia and some anxiolysis.
- iv. Contractility. (Must augment in conjunction with correction of acidosis, hypoxia, cessation of drugs such as beta-blockers). Inotropes classified as Catecholamines; Phosphodiesterase inhibitors (milrinone); cardiac glycosides; calcium; calcium sensitisers (Levosimendan). In acute heart failure, typically use Dobutamine or milrinone as they augment contractility as well as reducing LV afterload, and have a favourable effect on pulmonary vascular resistance. Remember that reducing afterload per se will often greatly improve cardiac output without affecting contractility.

As the patient deteriorates mechanical support of his circulation is considered. Describe techniques used for mechanical support of the circulation.

- a) Positive intrathoracic pressure (IPPV and PEEP either via ETT or facemask will reduce LV afterload. It often improves oxygenation and reduces work of breathing which can be a major O₂ requirement with pulmonary oedema. PEEP reduces atelectasis thereby reducing PVR and thus can improve RV function; over-distension of the lungs will worsen RV function.
- b) IABP. Placed percutaneously either in ITU, OR or ED. Balloon inflates with Helium at dichrotic notch (AV closure), and deflates immediately prior to LV ejection. Benefit twofold – increase diastolic BP and therefore LV perfusion, and also reduce LV afterload thus reducing work and improving CO.. Contraindicated with Aortic regurgitation. Risk of leg ischaemia. Timing critical – late deflation or early inflation increases cardiac work.
- c) LVAD. Bridge to recovery or transplant. Extremely expensive and invasive.
- d) ECMO. V-A ECMO for circulatory failure. Less invasive than LVAD; risk of limb ischaemia; bleeding;
- e) Surgery for acute valvular regurgitation or VSD, and to drain tamponade.

CASE TWO

Short case number: 4_06_02

Category: Infectious disease

Discipline: Intensive Care

Setting: Intensive Care

Topic: Sepsis_Septic Shock

CASE



A 65 year-old woman presents with a 3-day history of abdominal pain and vomiting. She has a history of hypertension, Type 2 Diabetes and obesity (BMI 32).

On examination she has a pulse rate of 110/bpm, BP of 85/45, temp of 39.5°C and seems confused. Her peripheries are warm. There is generalised abdominal tenderness and guarding.

CT scan shows an inflammatory mass involving rectum and adnexae.

At laparotomy, a diverticular abscess with faecal peritonitis is diagnosed and she undergoes a Hartmann's procedure and peritoneal washout.

She returns to the ICU intubated and passing minimal urine, and with a mean arterial blood pressure of 55-60mmHg.

QUESTIONS

1. What are Systemic Inflammatory Response Syndrome, Sepsis and Septic shock?
2. In assessing this patient outline the possible causes of sepsis and the pathophysiology that results in septic shock.
3. Outline the key clinical features that you would assess if you were concerned that this patient was in shock due to sepsis.
4. Outline the management of severe sepsis and septic shock.
5. As part of the management of this patient the team are considering commencing inotropic support. Give an overview of the vasoactive and inotropic agents that are often used and detail the different actions / effects on the circulation.

Resources:

- Berston, AD, & Soni, N (2009). Oh's intensive care manual (6th ed.). Philadelphia: Elsevier.
- Cameron et al (2009). Textbook of adult emergency medicine (3rd ed.). Sydney: Churchill Livingstone.

ANSWERS

What are Systemic Inflammatory Response Syndrome, Sepsis and Septic shock?

Sepsis, severe sepsis, septic shock a continuum. ACCP and SCCM consensus 1991. Sepsis – infection with systemic inflammation; severe sepsis = sepsis plus organ dysfunction; septic shock = sepsis with haemodynamic collapse. SIRS may or may not be associated with sepsis – may be caused by eg pancreatitis; cardiopulmonary bypass etc

In assessing this patient outline the possible causes of sepsis and the pathophysiology that results in septic shock.

Any microorganism or its products, classically bacteria but also virus, fungi eg candida; protozoans eg *P falciparum* malaria; bacterial products eg endo or exotoxins; Bacterial wall components bind to CD14 receptor on monocytes, also Toll-like receptors which activates intracellular processes leading to release of various cytokines eg TNF, IL1,2,6 – proinflammatory. Leads to vicious cycle of pro-coagulation, myocardial depression, microcirculatory dysfunction.

Outline the key clinical features that you would assess if you were concerned that this patient was in shock due to sepsis.

Shock refers to an inability to provide adequate blood flow and oxygen for tissue homeostasis. In septic shock, there is microcirculatory failure with vasodilation and “leaky” capillaries, and a high cardiac output. The patient often presents with hypotension and warm peripheries (as opposed to the “shut-down” appearance typical of hypovolaemic and cardiogenic shock). There is almost always fever but temperature must be measured in a “core” site eg nasopharynx; bladder; pulmonary artery; rectum. Axillary and oral temperature measurement is useless. There is typically a high cardiac output despite decreased myocardial contractility. Despite the high cardiac output there is impaired O₂ delivery to tissues leading to widespread organ dysfunction eg confusion; oliguria and rising Creatinine; hypoxaemia from lung dysfunction due to Acute Lung Injury (ALI) leading on to Acute Respiratory Distress Syndrome (ARDS). This continuum of respiratory dysfunction is not specific to sepsis, it simply reflects the limited range of responses to injury that are available to the lung. It consists of a neutrophil-rich exudate causing non-cardiogenic pulmonary oedema. Gut dysfunction is both cause and effect – gut ischaemia leads to bacteria entering the portal system; liver dysfunction may impair Kupffer cell function leading to systemic sepsis. Typical LFT abnormalities show an obstructive pattern due to intrahepatic cholestasis. Acalculous cholecystitis can also occur.

Outline the management of severe sepsis and septic shock.

Management has several arms – initial resuscitation; diagnosis; supportive treatment; specific treatment. This is a standard approach to critical illness and the various limbs of management must proceed concurrently.

i. Resuscitation. ABC. First 6 hours crucial and early goal-directed resuscitation is probably effective at reducing mortality. In septic shock there is usually intravascular volume depletion due to vasodilation and fluid loss into the interstitium. Fever also increases fluid requirements. Several litres may be required initially and despite a number of studies over many years attempting to show that colloids are “bad” there is no good evidence to support this. There is no doubt that excessive salt and water administration can exacerbate pulmonary dysfunction. If unable to keep MAP .65mmHg, add vasopressor, see below, initially noradrenaline. Hypoxaemia or obtundation may mandate intubation and mechanical ventilation. Temperature over 40C in the shocked patient should be managed aggressively including intubation, paralysis and active cooling. Be wary of paracetamol in the presence of liver dysfunction.

ii. Diagnosis of the cause of sepsis and of various organ dysfunctions. WBC count (neutrophilia suggests bacterial infection; lymphocytosis viral; neutropaenic sepsis requires a very broad-spectrum antibiotic approach;). Cultures of blood, wounds, pus, sputum, urine,; if no obvious source, CSF, throat,vagina, stool. Blood film for malaria if relevant. CXR. Elecs, Creat, urea, LFT.Ultrasound kidneys to rule out obstruction. CT abdomen if a possibility of pathology. Echocardiogram if a murmur is heard. As well as a complete history and physical examination, including a history of possible exposure.

iii. Specific treatment. This means antibiotics and source control. It may be as aggressive as amputation or major debridement for gas gangrene or other severe soft-tissue infections or as simple as removal of an infected vascular access device . Drainage of infected kidney; laparotomy as in this case;

Antibiotics initially are broad-spectrum and when cultures are available, they can be targeted more specifically. Initial spectrum chosen depending on likely source of sepsis, local known resistance patterns, comorbidities, presence of immunosuppression (eg febrile neutropaenic patient)

iv. Supportive treatment is a continuation of initial resuscitation and includes support of ventilation; circulation, usually with invasive haemodynamic monitoring with arterial, central venous and possibly pulmonary artery catheterisation; renal support, often with some form of CRRT.

v. Adjuvant therapy. No consensus. Activated Protein C – aim to reduce thrombin formation, endothelial cell apoptosis. May have a role. Steroids – have been in and out of favour on various occasions; currently recommended in low doses (2-300mg/day hydrocortisone) for those requiring vasopressors. Extracorporeal blood purification techniques - ?role?

Give an overview of the vasoactive and inotropic agents that are often used and detail the different actions / effects on the circulation.

vi.

Agent	Alpha1	Beta1	Beta2	Dopamine	CO	SVR
Adrenaline	+++	+++	++	0	++	+/0
Noradren	++++	+	+/0	0	0/+	+++
Dobutamine	0	+++	++	0	++	-
Dopamine	High dose	Mod dose	0	Low dose	+	+ - ++

Hence see that Dobutamine alone in septic shock will often lead to a fall in BP needing Noradrenaline to counteract.

No good evidence supporting one or other vasoactive drug combination over another. Important thing is to achieve MAP >60mmHg or so and monitor effects.

CASE THREE

Short case number: 4_06_03
Category: Renal and Urinary Systems
Discipline: Intensive Care
Setting: Intensive Care
Topic: Acute Renal Failure

CASE



A 50 year old man presents following an industrial accident where his legs were trapped in a masonry fall.

He sustained bilateral femoral fractures and major muscle damage.

Extrication was prolonged and there was a period of 2 hours with hypotension BP 70/40. He is now in the ICU 2 days later.

He is haemodynamically stable and afebrile but passing very little urine and his Creatinine is now 500micrograms/litre.

QUESTIONS

1. What are the key components of your assessment of the oliguria and why?
2. The team are concerned that the patient is developing acute renal failure. Define Acute Renal failure (ARF).
3. In the context of this clinical situation what are the possible causes of acute renal failure? Describe the effects of these possible causes on the kidney and renal function.
4. What are the clinical and biochemical features of ARF?
5. Outline the management of this oliguric patient.
6. What are the indications for, and general principle of, dialysis?

Resources:

- Berston, AD, & Soni, N (2009). Oh's intensive care manual (6th ed.). Philadelphia: Elsevier.
- Cameron et al (2009). Textbook of adult emergency medicine (3rd ed.). Sydney: Churchill Livingstone.

ANSWERS

The team are concerned that the patient is developing acute renal failure. Define Acute Renal failure (ARF).

Various definitions. . Occurs in 15-20% of ICU patients. RIFLE criteria (risk; injury; failure). Usually defined by rise in creatinine (indirect measure of GFR). Oliguria is less than 0.5ml/kg/hour urine. Best not to use 400ml/day as no account of body size.

In the context of this clinical situation what are the possible causes of acute renal failure? Describe the effects of these possible causes on the kidney and renal function.

Causes usually divided into Pre-renal, ie reduction in renal blood flow from any cause; Renal ie glomeruli/tubules/interstitium); and Post-renal ie obstruction at any level. Various biochemical indices have been used to distinguish pre-renal from renal causes but these have not been shown to be so useful. Eg Fractional excretion of sodium.

Important to recognize that many “Renal” causes of ARF are a continuum from severe and unrelieved pre-renal oliguria, and their separation serves little clinical purpose. Most of the remainder are due to drug toxicity particularly radiocontrast media and aminoglycosides. and cyclosporine. NSAID toxicity is probably best considered as an example of pre-renal due to effects on PG-mediated autoregulation. There are a few exceptions such as acute crescentic glomerulonephritis and acute interstitial nephritis. Important thing is that ARF is an independent predictor of death in ICU.

Because so many (>30%) of ICU patients have some degree of pre-existing renal impairment, these people have much less functional reserve so that a seemingly trivial insult can lead to acute-on-chronic renal failure.

What are the clinical and biochemical features of ARF?

The hallmark is a rising Creatinine and Urea. Creatinine rises curvilinearly as GFR falls; GFR is around 50% or less of normal before any increase in P creat. Urea depends on many other factors – liver function, hydration, corticosteroids etc . Symptomatic uraemia in adults with urea >30mmol/l, maybe lower (encephalopathy, itch, pericarditis etc) Hyperkalaemia, Metabolic acidosis, Hyperphosphataemia. Majority are oliguric; can have a “polyuric” phase during recovery from intrinsic renal failure. Non-oliguric ARF (whether urine output is diuretic-induced or not) may make dialysis less necessary and certainly simplifies fluid management. Other clinical manifestations include impaired platelet function.

Outline the management of this oliguric patient.

Management involves aggressive resuscitation and monitoring of urine output and haemodynamics, while removing the cause. Once ARF is established, management revolves around treatment or removal of the cause and maintenance of physiologic homeostasis while awaiting recovery. Supportive care includes nutrition (which requires modification in ARF), prevention of infection, and prevention of major electrolyte and acid-base disturbance which should never occur in a modern ICU. Treating the cause of ARF will include imaging to rule out obstruction,

What are the indications for, and general principle of, dialysis?
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Dialysis. Indications. Symptomatic uraemia (difficult to define precisely); severe hyperkalaemia or acidosis; fluid overload; pericarditis; There are no RCT's that define the ideal time to commence Renal replacement therapy.

CASE FOUR

Short case number: 4_06_04

Category: Gastrointestinal and Hepatobiliary Systems

Discipline: Emergency Medicine & Intensive Care

Setting: Emergency Department / Intensive Care

Topic: Acute Hepatic Failure

CASE



You are the Intensive Care Resident Medical Officer on night shift. At 11pm your Registrar informs you that the Unit is receiving a new admission from the Emergency Department. She asks you to go to Emergency "to find out what the story is".

22 year old Tracey Sharpe has presented to the Emergency Department, stating she has taken a large quantity of paracetamol in a suicide attempt. She presented at 8pm this evening. You go to the bedside and take a history from Tracey.

She States took overdose of the following between 2pm and 5pm on the previous day (30hours ago) which included:

Paracetamol: 96 tablets of 500mg = approximately 60gm total = 1200mg/kg

Ibuprofen: 12 tablets of 200mg = 2400mg = approximately 50mg/kg

Tracey States she wanted to kill herself. She also admits to also having alcohol with the tablets. States she fell asleep after taking the pills and alcohol, awoke at 9pm last evening, and felt OK so went back to bed.

Today she has presented complaining of abdominal pain, RUQ in location, dull ache, severity 5/10. She is also complaining of nausea, and has had 2x vomits while in the Emergency Department.

She has a background history of having a previous paracetamol overdose following the breakup of her relationship 3 years ago, and depression. She is not taking any medications and has no allergies.

While you are with the patient you notice a print out of her blood results:

EUC: Na 134, K 4.5, CL 102, Bicarb 18, Urea 3.1, Cr 53

LFT: Bilirubin 41, Albumin 34, Alk Phos 119, GGT 104, ALT 3255, AST 8589

CMP: Calcium Corrected 2.19, PO4 0.83, Mg 0.82

Paracetamol Level: 339

Arterial Blood Gas: pH 7.310, PCO2 34, PO2 103, Bicarb 17, BE -8.4

FBC: WCC 23.9, Hb 142, Platelets 259

COAGS: PT 39, INR 3.6, APTT 28 sec

QUESTIONS

1. What would be the key components of your assessment of Tracy and why?
2. Outline the management of patients presenting with Paracetamol overdose.
3. Tracey is reluctant to stay in hospital. You advise Tracey that it is likely that the Paracetamol has significantly damaged her liver. In your explanation to Tracey as to why she should stay revise the functions of the liver and how they are affected in liver failure due to paracetamol toxicity.
4. Explain acute hepatic failure and outline the common causes
5. What is the clinical picture of fulminant hepatic failure and explain the underlying pathophysiology?
6. Outline the key principles of management of acute liver failure.
7. Tracey asks whether she can 'get a new liver'. Revise and explain the indications for liver transplantation. Is Tracey eligible to be placed on the organ transplant register?

Resources:

- Berston, AD, & Soni, N (2009). *OH's intensive care manual* (6th ed.). Philadelphia: Elsevier.
- Cameron et al (2009). *Textbook of adult emergency medicine* (3rd ed.). Sydney: Churchill Livingstone.

Electronic Resources:

- Daly, F, Fountain, JS, Murray, L, Graudins, A, Buckley, NA (2008). Consensus statement: guidelines for the management of paracetamol poisoning in Australia and new Zealand – explanation and elaboration. *MJA*, 188(5), 3 March 2008, p296-301.

ANSWERS

What would be the key components of your assessment of Tracy and why?

Coagulopathy, Jaundice and encephalopathy developing within days to weeks of an insult.
Heterogeneous condition – due to varying age and co-morbidities, varying aetiologies, and varying duration of evolution.

Outline the management of patients presenting with Paracetamol overdose.

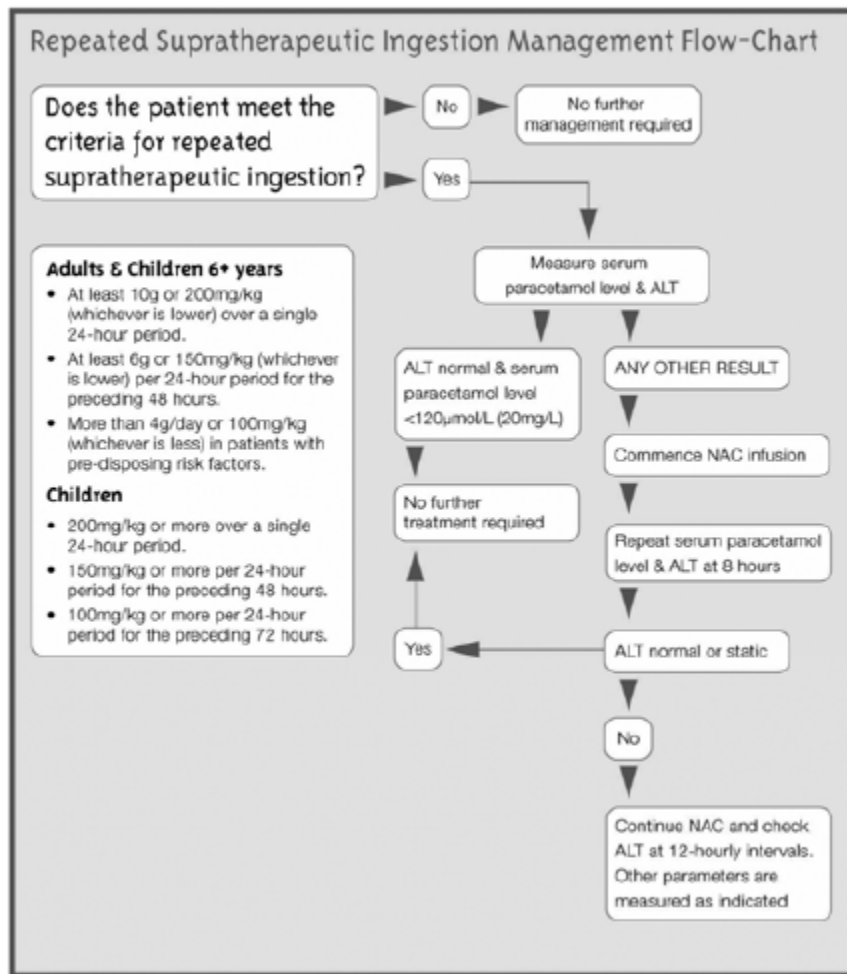
Refer to Paracetamol Guidelines, Electronic Resources

3 Management flow chart for acute paracetamol exposure with known time of ingestion



ALT = alanine aminotransferase. NAC = N-acetylcysteine. OD = overdose.
This flow chart is applicable only if the treating clinician is confident of an accurate time of ingestion. For management of sustained-release product overdose, see "Sustained-release paracetamol overdose" section in text. ♦

4 Management flow chart for repeated supratherapeutic paracetamol ingestion



ALT = alanine aminotransferase. NAC = N-acetylcysteine. ♦

In your explanation to Tracey as to why she should stay revise the functions of the liver and how they are affected in liver failure due to paracetamol toxicity.

O'Grady definitions

i.hyperacute : onset of encephalopathy within 7 days of jaundice. Commonly paracetamol, ischaemia, viral, toxins eg amanita. Best chance of spontaneous recovery but also greatest degree of cerebral oedema, encephalopathy, coagulopathy.

iiAcute. Encephalopathy 8-28 days after jaundice

iiiSubacute. Encephalopathy >28days after jaundice. Least chance of recovery. More often idiopathic and drug-related. May be difficult to distinguish from chronic disease as can present with ascites.

Explain acute hepatic failure and outline the common causes

Aetiologies changing – eg less HBV since widespread immunisation but still common esp in France. Viral 40-70% worldwide. Up to 10% from HAV although an individual with Hep A is very unlikely to develop ALF but it is a very common disease worldwide. HCV rarely causes ALF, usually CLD instead. 15-25% drug-induced, mostly paracetamol, also rarely ecstasy. Acute Wilson's disease, HELLP (pregnancy-associated); Budd-chiari (venous obstruction) and ischaemia uncommon.

The aetiology is important, both for therapeutic and prognostic reasons. Thorough history, serologic testing, autoantibody screen, drug assays, etc vital

- 1) Liver functions include :
 - i. Intermediary metabolism. Carbohydrate, esp as "Glucostat. Protein, Lipid. Gluconeogenesis, Glycogen storage, etc. Ammonia and Urea from Amino acid metabolism.
 - ii. Protein synthesis, eg coagulation factors, albumin, transport proteins etc
 - iii. Bilirubin metabolism and formation of bile
 - iv. Immune – Kupffer cells clear microorganisms that enter portal system before they enter systemic circulation
 - v. Drug and toxin metabolism by various pathways eg hydroxylation, oxidation, glucuronidation
 - vi. Hormone manufacture

Consequences of liver failure will include derangements of some or all of these functions eg Jaundice, encephalopathy from NH₄; hypoglycaemia; coagulopathy; hypoalbuminaemia etc. The magnitude of each will partly depend on the duration of the ALF.

What is the clinical picture of fulminant hepatic failure and explain the underlying pathophysiology?

In particular, ALF is associated with **encephalopathy** from cerebral oedema as well as other probable causes including NH₄ accumulation leading to intracerebral GLN leading to oedema. Can progress with astonishing rapidity. Graded 0 – 4 with 0 = normal, to 4 = coma +/- decerebrate. More severe and more rapid onset with hyperacute; whereas in CLD there is adaptation and control of intracellular osmolality (?myoinositol?) so oedema rare and of low grade; **coagulopathy**. The hallmark Also consumptive thrombocytopaenia common; **sepsis** due to impaired cell-mediated immunity, impaired phagocytosis and decreased complement levels. Also Kupffer cell function impaired. **hypotension** -Impaired circulatory control - hyperdynamic with vasodilation and high CO. May have impaired ACTH response. Renal failure common -50% **Hypoglycaemia**

Outline the key principles of management of acute liver failure.

- i. ABC. Arrange transfer to specialised unit.
- ii. Determine the cause. Specific treatment if indicated. Especially n-Acetyl Cysteine if Paracetamol poisoning.. Follow readily-available time/concentration nomogram for applicability. Antiviral drugs may prevent recurrence in viral aetiologies.

iii. Management involves providing an environment most conducive to liver regeneration, or stability until a transplant is available (see 5.) extracorporeal liver support remains experimental

iv. Ventilation if indicated – all grade 3 – 4 encephalopathy; ICP control with jugular oximetry and ICP measurement. Invasive haemodynamic monitoring. Prevention of hypoglycaemia. Meticulous prevention and management of sepsis. Renal replacement therapy early rather than late, particularly to control acidosis. Remember urea may not rise much due to failure of hepatic protein metabolism. Coagulation support before procedures and if bleeding, otherwise not, as PT is useful prognostically.

Tracey asks whether she can 'get a new liver'. Revise and explain the indications for liver transplantation. Is Tracey eligible to be placed on the organ transplant register?

Use King's College criteria for non-survival. Paracetamol-induced : pH <7.3 following volume resuscitation if >24hours post-ingestion; OR PT>100secs AND creat>300 AND grade 3-4 encephalopathy. Non-paracetamol : PT>100; OR pH <7.3 OR encephalopathy PLUS ANY 3 OF : i. Age <10 or >40; ii. nonAnonB or drug-induced; iii. Jaundice-to-encephalopathy > 7days; iv.PT>50secs; v. Bilirubin>300.

Remember that there have been improvements in supportive care including ECLS since these criteria were promulgated but they still hold as prognostic criteria. Transplantation is riskier in FHF because patients' sicker, marginal donors used